MAST20009 Vector Calculus

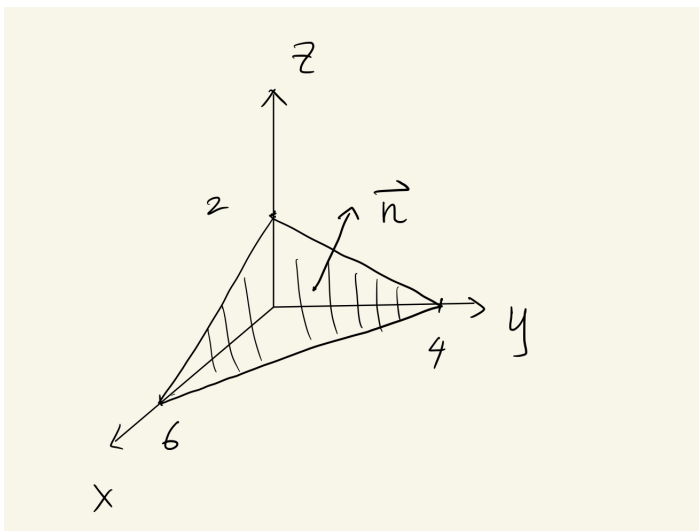
Problem Sheet 5: Integrals over Surfaces

Worked Solutions for selected questions

Q89. Solution:

$$\iint_S \boldsymbol{F} \cdot d\boldsymbol{S}, \qquad \boldsymbol{F} = 18z\mathbf{i} - 12\,\mathbf{j} + 3y\mathbf{k}$$

$$S\colon \quad 2x + 3y + 6z = 12, \qquad x, y, z \geq 0, \qquad \boldsymbol{n} \quad \text{upwards}$$



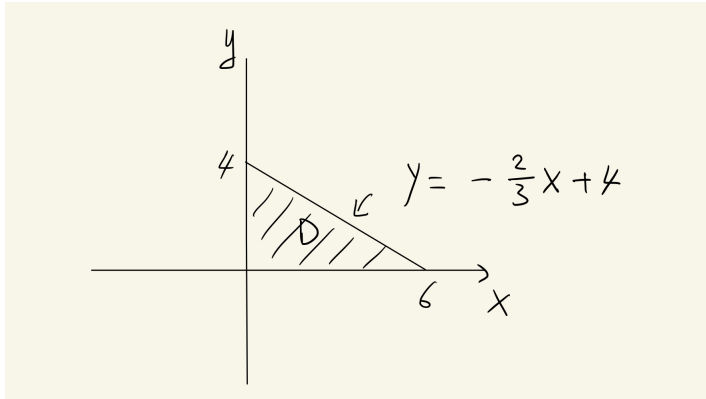
The upward normal to the surface S is $\boldsymbol{n} = (2, 3, 6)$

$$|\boldsymbol{n}| = \sqrt{4 + 9 + 36} = \sqrt{49} = 7 \Rightarrow \hat{\boldsymbol{n}} = \frac{1}{7}\,(2, 3, 6) \Rightarrow |\hat{\boldsymbol{n}} \cdot \mathbf{k}| = \frac{6}{7}$$

Hence the surface integral:

$$\begin{aligned}
\iint_S \boldsymbol{F} \cdot d\boldsymbol{S} &= \iint_S (8z\mathbf{i} - 12\,\mathbf{j} + 3y\mathbf{k}) \cdot \frac{1}{7}\,(2, 3, 6)\; dS = \frac{1}{7} \iint_S (36z - 36 + 18y)\, dS \\
&= \frac{18}{7} \iint_S (2z - 2 + y)\, dS = \frac{18}{7} \iint_D \frac{4 - \frac{2x}{3} - y - 2 + y}{|\hat{\boldsymbol{n}} \cdot \mathbf{k}|}\, dx\, dy \\
&= \frac{18}{7} \cdot \frac{7}{6} \iint_D (2 - \frac{2x}{3})\, dx\, dy = \iint_D (6 - 2x)\, dx\, dy
\end{aligned}$$

where D is the projection of S in xy-plane:



Using vertical strips: $D = \{(x, y) \in \mathbb{R}^2 \mid 0 \le x \le 6, \quad 0 \le y \le 4 - \frac{2x}{3}\}$.

Hence the surface integral:

$$\begin{aligned}
\iint_S \boldsymbol{F} \cdot d\boldsymbol{S} &= \int_0^6 \int_0^{4-\frac{2x}{3}} (6 - 2x)\, dy\, dx = \int_0^6 [6y - 2xy]_{y=0}^{y=4-\frac{2x}{3}}\, dx \\
&= \int_0^6 24 - 4x - 8x + \frac{4x^2}{3}\, dx = \int_0^6 24 - 12x + \frac{4x^2}{3}\, dx \\
&= \left[24x - 6x^2 + \frac{4}{9}x^3\right]_{x=0}^{x=6} = 24(6) - 6(36) + \frac{4}{9}(216) = 24
\end{aligned}$$

Method 2: Parametrise S by

$$x = u, \quad y = v, \quad z = 2 - \frac{u}{3} - \frac{v}{2}.$$